Experiment 6

Energy Calibration in Scintillation Detectors

Objective:

1- To find the energy calibration by using two radioactive sources.
2- Calculate the energy of unknown source

Theory:

Energy calibration means evaluating a relationship between the energy of the emitted particle and the channel number of the peak. Once the energy calibration is known, it is easy to find the energy of unknown source. The energy proportional to the channel number of the peak, that is:

$$E = C \times CH \qquad (1)$$

where, C is a constant.

If we draw a relation between the energy and the channel number of the peak, then the slope of the line must equal to the constant C, that is:

$$E_0 = \text{slope} \times CH_0 \qquad (2)$$

Apparatus:

Scintillation detector.

Computer.

Two known radioactive sources.

Unknown radioactive source

Procedure:

1- Connect the detector to computer.
2- Put Gamma source inside the detector.
3- From **start** in computer go to **programs menu** and then go to **spectech** (spectrum technique).
4- Change the setting of the **700 keV** to **on**.
5- Go to **mode** choice **start** to start detecting radiations.
6- After 15 min go to **mode** and press **stop**.
7- By right click on mouse and choose set **ROI** and then shade the area under the peak.
8- Go to **setting** choice **energy calibration** and then choose **2 points**.
9- Move the pointer to the maximum count of the peak.
10- Record the value of channel number (CH_0) at the maximum counts and the energy calibration (E_0).
11- Repeat the same steps for the second source and unknown source.
12- Plot a graph between CH_0 and E_0 to find the slope.
13- Calculate the energy E_0 of unknown source.

Results:

Source description

Element	Activity (A_0) (μCi)	Half life ($t_{1/2}$) (year)

source	CH_0	E_0 (keV)

C = Slope = ……………….

E_0 = Slope × $(CH_0)_{unknown}$ = ……………… keV.

E_0 = ……………keV.   (From Fig.1)

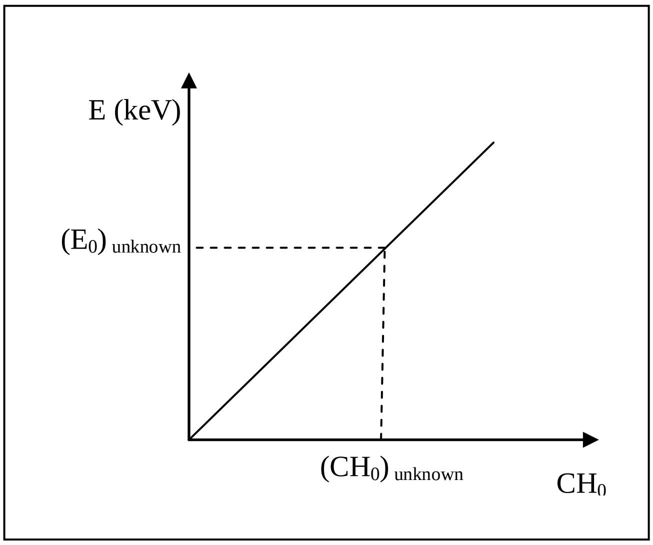

Fig.1